INCH-POUND

MIL-S-22885/90B
23 June 1989
SUPERSEDING
MIL-S-22885/90A
29 July 1981

MILITARY SPECIFICATION SHEET

# SWITCHES, PUSHBUTTON, ILLUMINATED, Ⓑ 4-LAMP, 0.750 SQUARE, LOW LEVEL TO 7 AMPERES, (DRIPPROOF, EMI/RFI SHIELDING, NIGHT VISION GOGGLE COMPATIBLE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-22885

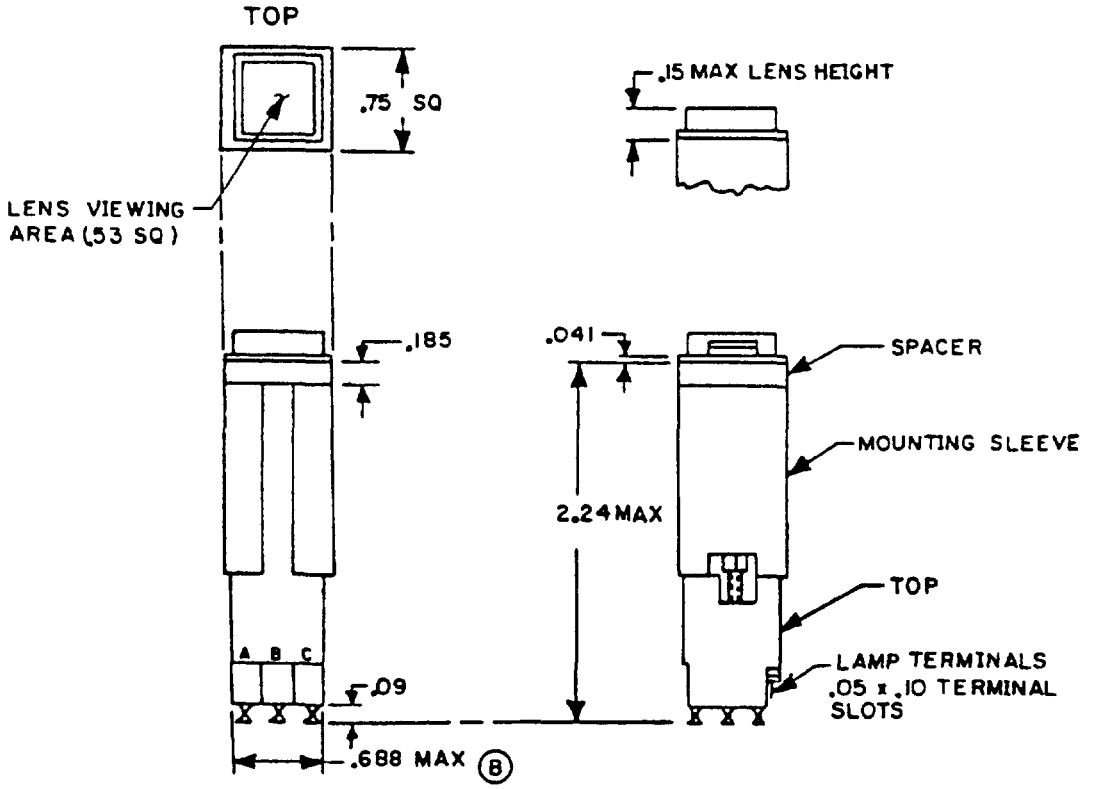



FIGURE 1. Switch, individual mounted (type I).

Ⓑ denotes changes

AMSC N/A FSC 5930

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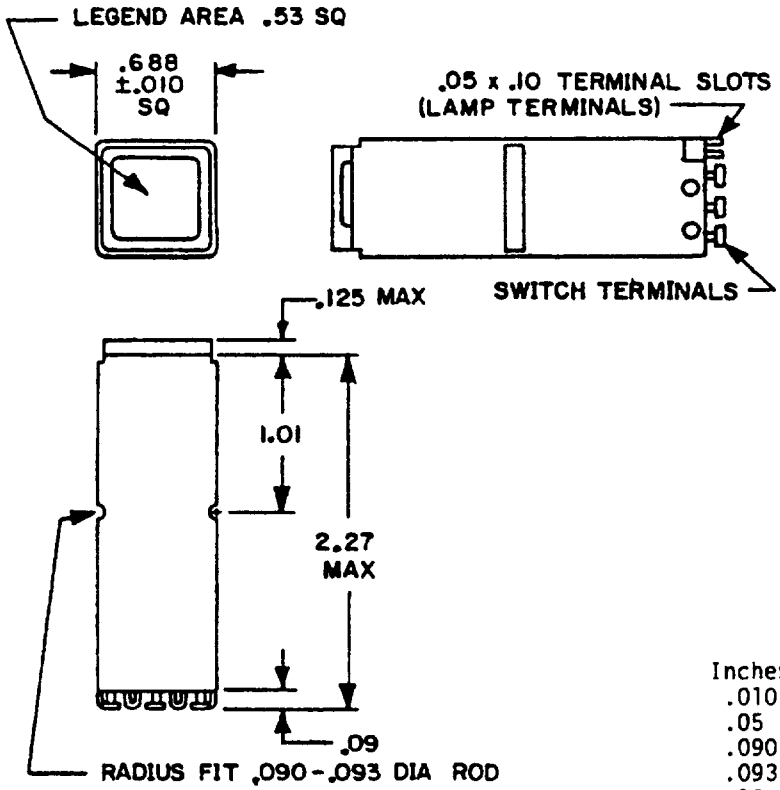
| Inches | mm |
|---|---|
| .041 | 1.04 |
| .05 | 1.3 |
| .09 | 2.3 |
| .10 | 2.5 |
| .15 | 3.8 |
| .185 | 4.70 |
| .53 | 13.5 |
| .688 | 17.48 |
| .75 | 19.0 |
| 2.24 | 56.9 |

NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
(B) 3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Unit mounts in panels from .031 to .19 (0.79 to 4.8 mm) thick.
5. A spacer is supplied with each switch unit for MIL-P-7788 panel requirements. Spacer places switch mounting flange flush with top of overlay panel.
6. Spacer to be used only for edge-light panels, discard otherwise.
7. Mounting sleeve shall not exceed .760 (19.30 mm) square and shall be reversible to accommodate edge-light panel spacer requirements.
8. Pushbutton is designed to prevent incorrect insertion into switch housing.
9. Exact shape of switch optional provided dimensions specified are not exceeded.
(B) 10. Terminal dimensions are not applicable to printed circuit board terminations.

FIGURE 1 Switch, individual mounted (type I) - Continued



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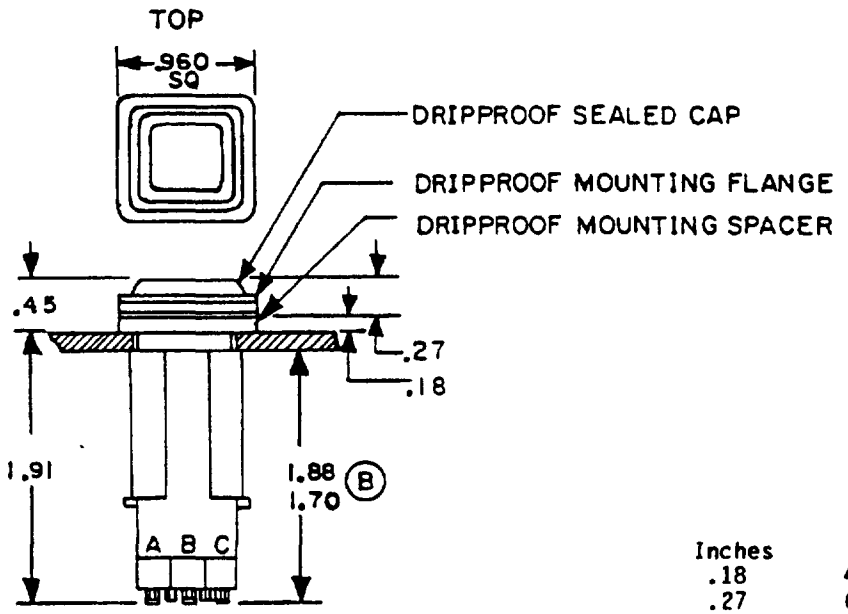

| Inches | mm |
|---|---|
| .010 | 0.25 |
| .05 | 1.3 |
| .090 | 2.29 |
| .093 | 2.36 |
| 10 | 2.5 |
| .125 | 3.18 |
| .53 | 13.5 |
| .688 | 17.48 |
| 1.01 | 25.7 |
| 2.27 | 57.7 |

NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
(B) 3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Pushbutton is designed to prevent incorrect insertion into switch housing.
5. Exact shape of switch optional provided dimensions specified are not exceeded.
(B) 6. Terminal dimensions are not applicable to printed circuit board terminations.

FIGURE 2. Switch, matrix mounted (type II).



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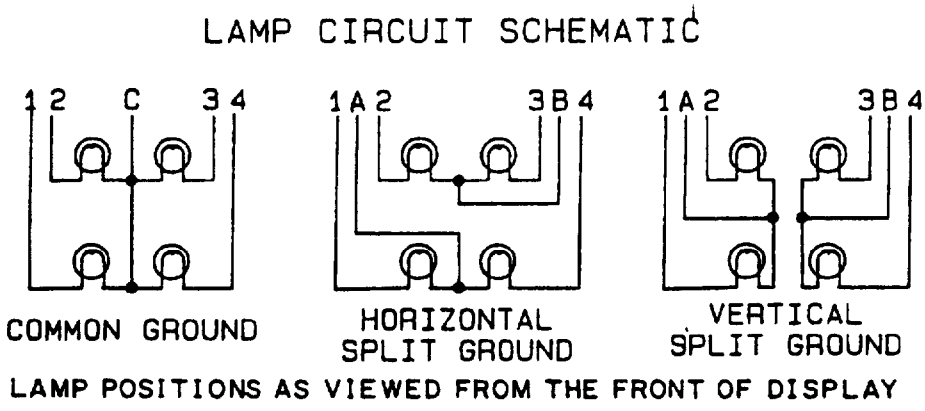

| Inches | mm |
|---|---|
| .18 | 4.6 |
| .27 | 6.9 |
| .45 | 11.4 |
| .960 | 24.38 |
| 1.72 | 43.7 |
| 1.88 | 47.8 |
| 1.91 | 48.5 |

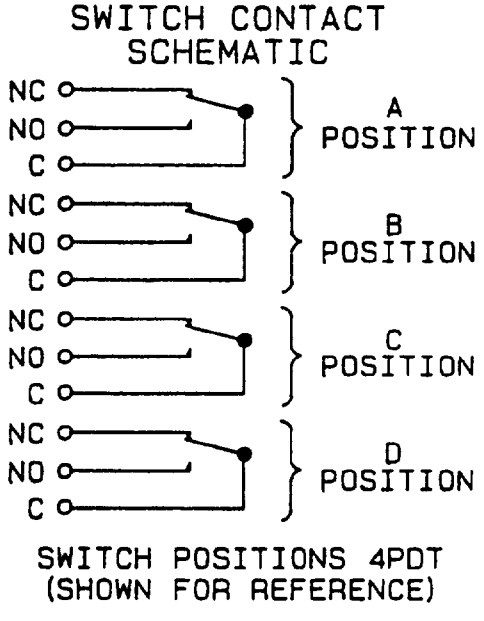
NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
(B) 3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Unit mounts in panels from .031 to .19 (0.79 to 4.8 mm) thick.
5. A spacer is supplied with each switch unit for MIL-P-7788 panel requirements.
6. Spacer to be used only for edge-light panels, discard otherwise.
7. Mounting sleeve shall not exceed .760 (19.30 mm) square and shall be reversible to accommodate edge-light panel spacer requirements.
8. Pushbutton is designed to prevent incorrect insertion into switch housing.
9. Exact shape of switch optional provided dimensions specified are not exceeded.
(B) 10. Terminal dimensions are not applicable to printed circuit board terminations.

FIGURE 3. Switch, dripproof sealed (type III).



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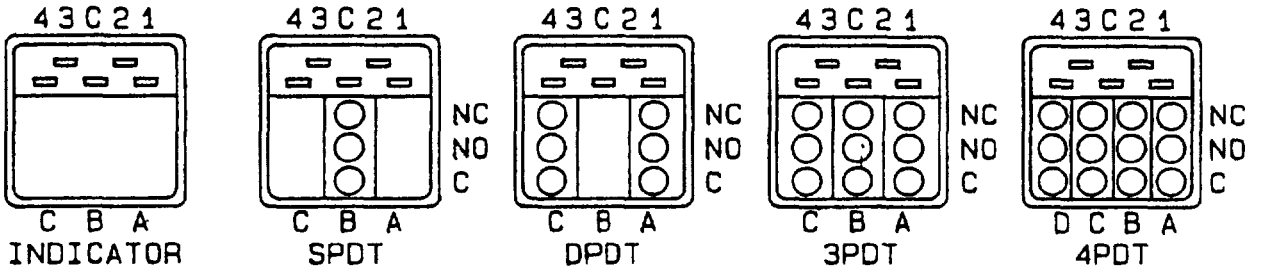

(B) FIGURE 4. Switch and lamp circuit schematics and termination arrangements and identification.



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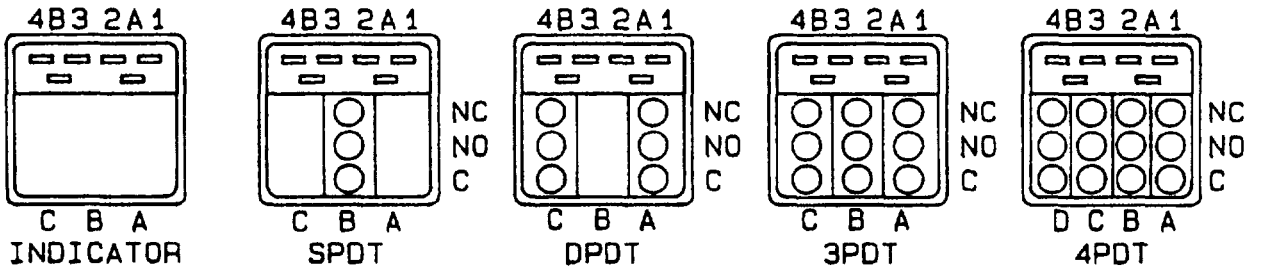
COMMON GROUND, SOLDER TERMINATIONS - REAR VIEWS

TOP



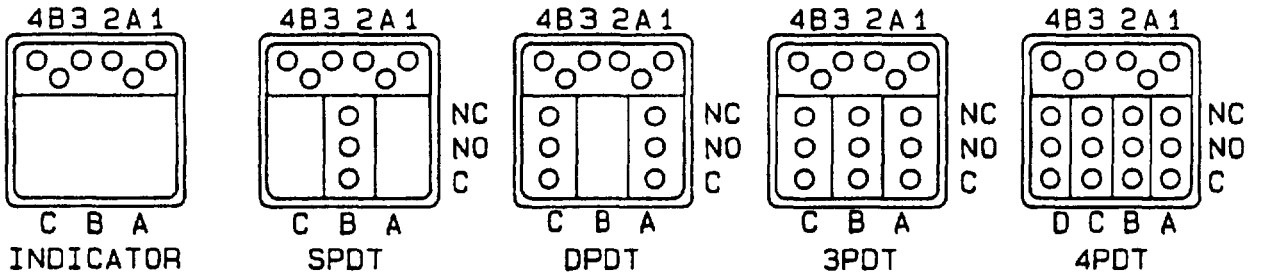

SPLIT GROUND, SOLDER TERMINATIONS - REAR VIEWS

TOP




SPLIT GROUND, PC BOARD TERMINATIONS - REAR VIEWS

TOP

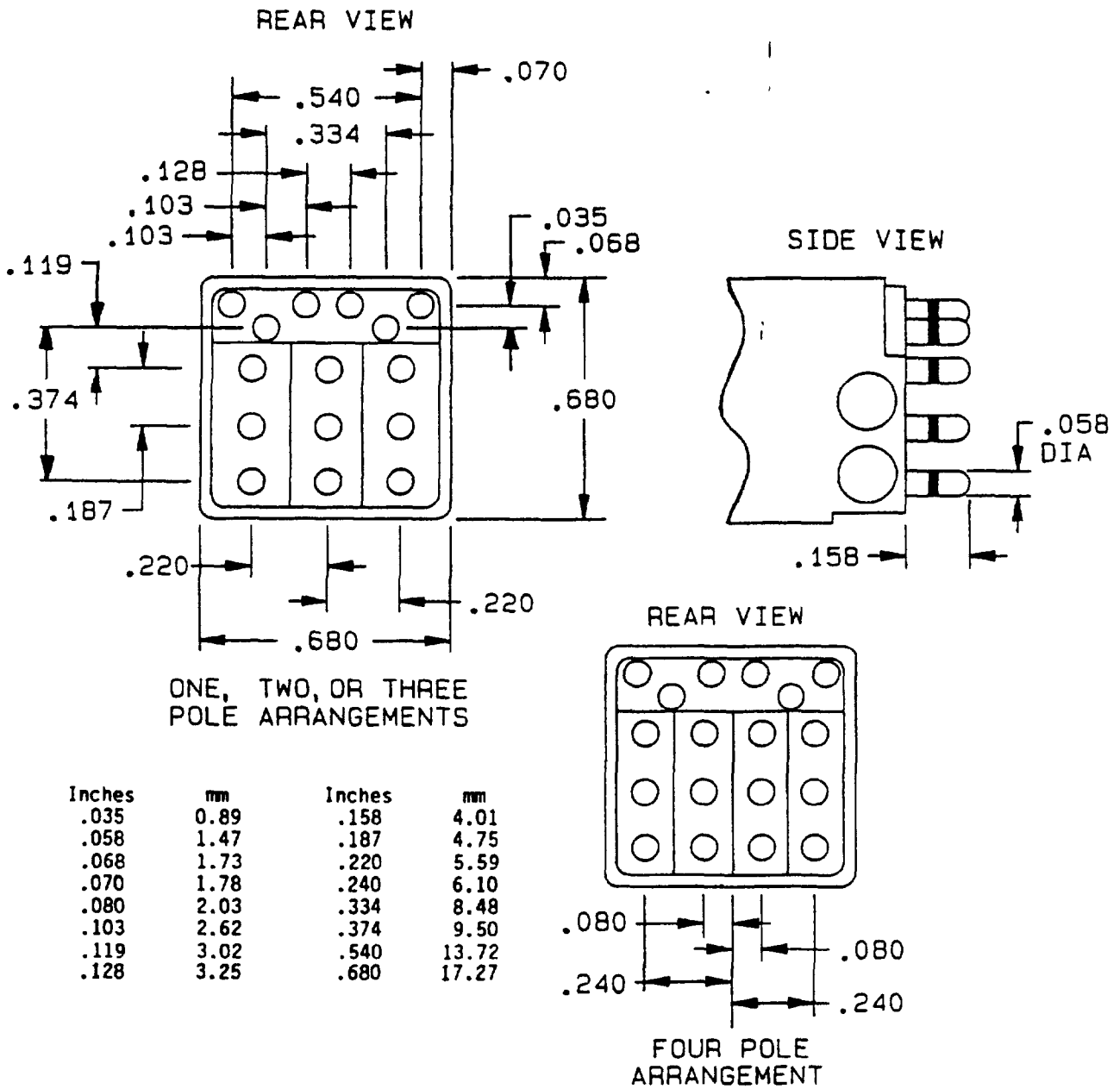



NOTE: Terminals and basic switch identification shall be permanently marked as shown.

Ⓑ FIGURE 4. Switch and lamp circuit schematics and termination arrangements and identification - Continued



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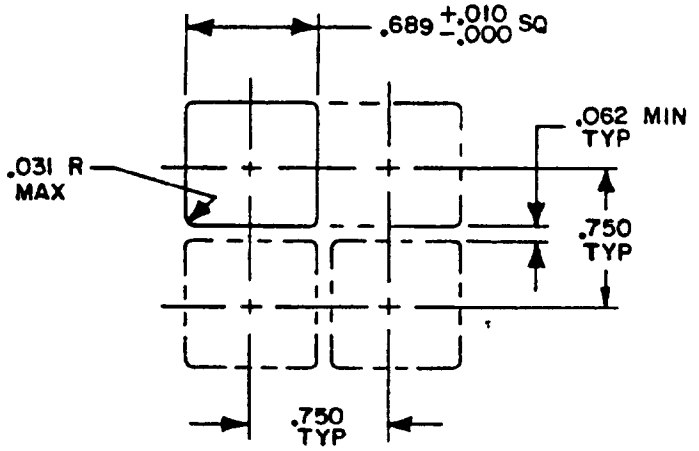

| Inches | mm | Inches | mm |
|---|---|---|---|
| .035 | 0.89 | .158 | 4.01 |
| .058 | 1.47 | .187 | 4.75 |
| .068 | 1.73 | .220 | 5.59 |
| .070 | 1.78 | .240 | 6.10 |
| .080 | 2.03 | .334 | 8.48 |
| .103 | 2.62 | .374 | 9.50 |
| .119 | 3.02 | .540 | 13.72 |
| .128 | 3.25 | .680 | 17.27 |

NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal positions and dimensions for four pole arrangements are the same as for three pole arrangements except as specified.

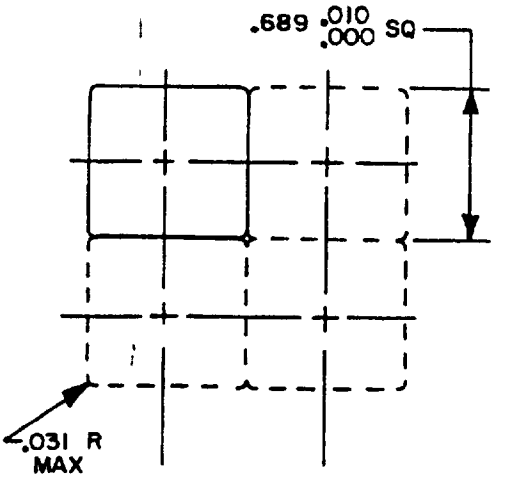
(B) FIGURE 5. Lamp and switch printed circuit pin terminal positions and dimensions.



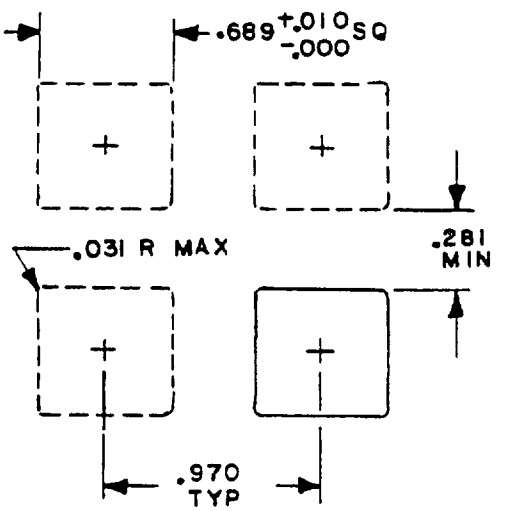
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RECOMMENDED PANEL CUTOUT
FOR INDIVIDUAL MOUNTING
(TYPE I SWITCH)




RECOMMENDED PANEL CUTOUT
FOR MATRIX MOUNTING
(TYPE II SWITCH)




RECOMMENDED PANEL CUTOUT
FOR DRIPPROOF SEALED
(TYPE III SWITCH)

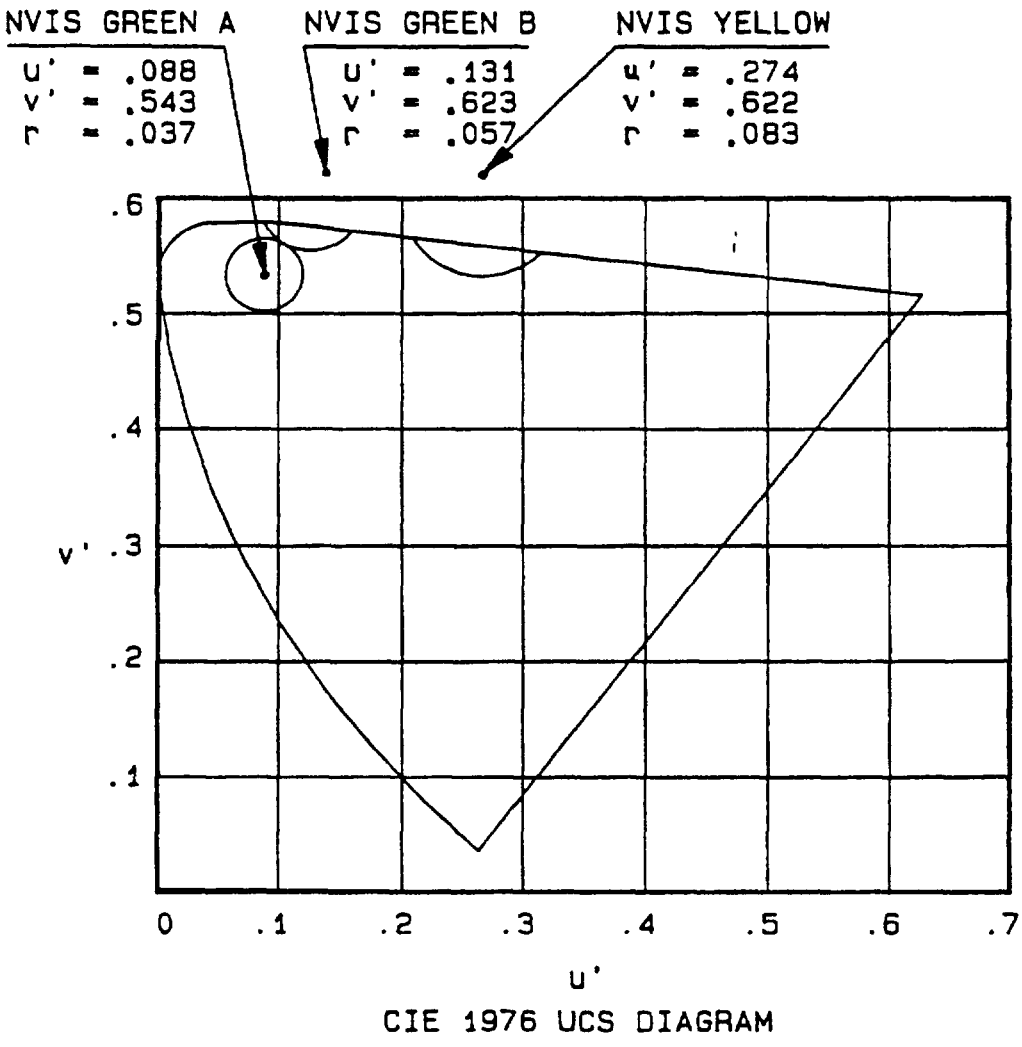
| Inches | mm |
|---|---|
| .010 | 0.25 |
| .031 | 0.79 |
| .062 | 1.57 |
| .281 | 7.14 |
| .689 | 17.50 |
| .750 | 19.05 |
| .970 | 24.64 |

NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.

FIGURE 6. Recommended panel cutouts.



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CIE 1976 UCS DIAGRAM

(B) FIGURE 7. Color limits for NVIS night vision goggle compatible colors.



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REQUIREMENTS:

Ⓑ Design and construction: See figures 1 through 6.

Ⓑ Complete switch shall consist of.

1 - Switch body: Includes MIL-S-8805/4 or /109 basic switches listed on QPL-8805. Type I bodies also include mounting sleeve and panel spacer. Type III bodies also include mounting sleeve, dripproof mounting flange and panel mounting spacer. EMI/RFI shielding is available when specified (see Part or Identifying Number (PIN)).

1 - Pushbutton: Includes display and lampholder. When specified, EMI/RFI shielding and night vision goggle compatibility are available (see PIN).

4 - Lamps: T-1 submidget flanged base incandescent. Lamps are not supplied; order separately.

Enclosure design. 1 (unsealed) for types I and II.
2 (dripproof) for type III.

Temperature characteristic: 1 (-55°C to +85°C for types I and II).
(-48°C to +85°C for type III).

Vibration grade: 2 (10-500 Hz).

Ⓑ Display type: S (sunlight readable, specularly reflective condition).
N (opaque background, visible legend).

Ⓑ Material.

Housing: Corrosion resistant steel.

Housing bezel: Corrosion resistant steel, black.

Panel mounting spacer: Corrosion resistant steel, black, for type I. Thermoplastic, nylon, black, for type III.

Front lens: High temperature heat resistant thermoplastic.

Operating characteristics:

Actuation force: 2 to 5 pounds.

Actuation travel: .150 ±.031 inch.

Lens module extraction force: 2 to 5 pounds.

Moisture resistance: Following step 6 of the moisture resistance test, the insulation resistance measurement shall be not less than 5 megohms.

Ⓑ Dielectric withstanding voltage at reduced barometric pressure: Applicable.

Ⓑ Weight: 1.6 ounces maximum.

Contact resistance: Not applicable after electrical endurance, inductive load dc.

Electrical ratings: See table I.

Low level life: Applicable to gold contact switches, 25,000 cycles.



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Ⓑ Color and luminance: See tables II and III for display type S.
See tables IV and V for display type N.

Ⓑ Sunlight readability (display type S):

Requirements: All legends shall be capable of being read in direct sunlight and at any glare producing, specularly reflective angles up to 15° ±2°C to the normal of the display viewing surface. The minimum average luminance contrast ratio of each lighted legend character to background shall be 0.6 minimum, except 0.4 minimum for blue and NVIS compatible colors. The average luminance contrast ratio of each unlighted legend character to background shall have an absolute value equal to or less than 0.1. The minimum average luminance contrast ratio for each color shall not be less than the values listed in tables VI and VII for the respective glare producing angle.

Test method: The test procedure for measuring the average luminance contrast ratios for sunlight readability shall be the specular reflectance test in accordance with MIL-S-22885.

Ⓑ Visible legend (display type N):

Requirements: All legends shall be visible white with an opaque background. The legend characters shall always be visible in any light ambience except in darkened conditions. In darkened conditions the legend characters shall illuminate in color with an average luminance of 0.5 to 3.0 foot-lamberts when energized at full rated voltage. See tables IV and V for specific colors and luminance.

Test method: The test procedure for measuring the average luminance and color of the illuminated visible white legend characters shall be in accordance with MIL-S-22885.

Ⓑ Night vision goggle (NVG) compatibility:

Requirements: The chrominance and luminance of the legend characters shall be compatible with NVIS (Night Vision Imaging System) goggles as specified in MIL-L-85762. The NVIS Green A color shall be within a circle where $u'_1$ = .088, $v'_1$ = .543 and the radius = .037 when the legend luminance is set at 0.1 foot-lamberts.. The NVIS Green B color shall be within a circle where $u'_1$ = .131, $v'_1$ = .623 and the radius = .057 when the legend luminance is set at 0.1 foot-lamberts. The NVIS Yellow color shall be within a circle where $u'_1$ = .274, $v'_1$ = .622 and the radius = .083 when the legend luminance is set at 15.0 foot-lamberts. Formula I is solved for the radius r to confirm compliance for any of the three colors. See figure 7 for NVIS color limits.

The spectral radiance of the legend characters shall be compatible with NVIS goggles as specified in MIL-L-85762 where NVIS radiance (NR) is defined. The NVIS Green A spectral radiance output shall not exceed $1.7 \times 10^{-10}$ NR for type I, class A and shall not exceed $1.6 \times 10^{-10}$ NR for type II, class B goggles when the legend luminance is set at 0.1 foot-lamberts. The NVIS Green B spectral luminance shall not exceed $1.7 \times 10^{-10}$ NR for type I, class A and shall not exceed $1.6 \times 10^{-10}$ NR for type II, class B goggles when the legend luminance is set at 0.1 foot-lamberts. The NVIS Yellow spectral radiance output shall not exceed $1.5 \times 10^{-7}$ NR nor be less than $5.0 \times 10^{-8}$ NR for type I, class A goggles, and shall not exceed $1.4 \times 10^{-7}$ NR for type II, class B goggles when the legend luminance is set at 15.0 foot-lamberts.



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Test methods. The text procedure for measuring the NVIS Green A, NVIS Green B or the NVIS Yellow luminance and color shall be in accordance with MIL-L-85762. The test procedure for measuring the spectral radiance of the legend for either color shall be in accordance with MIL-L-85762.

$$(u' - u'_1)^2 + (v' - v'_1)^2 \leq r^2 \qquad \text{Formula I}$$

where:

$u'$, $v'$ - - - - = 1976 UCS chromaticity coordinates of the test article.
$u'_1$, $v'_1$ - - - = 1976 UCS chromaticity coordinates of the center point of the specified color area above.
r- - - - - - - = radius of the allowable circular area on the 1976 UCS chromaticity diagram for the specified color.

When type S (sunlight readable) displays are required to be night vision goggle compatible, they shall meet the MIL-L-85762 minimum contrast requirement of 0.4 at rated voltage. Colors for type S (sunlight readable) night vision goggle compatible displays shall be NVIS Green B and NVIS Yellow (see table III). Colors for the type N (visible white) night vision goggle compatible displays shall be NVIS Green A and NVIS Green B (see table V).

Ⓑ TABLE I. Electrical ratings.

| Contact material | Load | Sea level | | 50,000 feet 28 V dc |
|---|---|---|---|---|
| | | 28 V dc | 115 V ac, 60 Hz | |
| | | (amperes) | (amperes) | (amperes) |
| Silver | Resistive | 7 | 7 | 4 |
| | Inductive | 4 | 7 | 2.5 |
| | Lamp | 2.5 | 2 | 2.5 |
| Gold | Resistive | 1 | --- | 1 |
| | Inductive | 0.5 | --- | 0.5 |
| | Low level life applicable. 30 millivolts maximum or peak ac at 10 milliamperes maximum. | | | |

Ⓑ EMI shielding (when applicable):

Requirement: The EMI/RFI shielding attenuation shall be not less than 60 dB over the frequency range from 100 to 1000 MHz.

Test method: Switches shall be tested to determine the shielding effectiveness in accordance with MIL-S-22885.



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Measurements are to be performed at the following frequencies:

| Frequency | Minimum attenuation |
|---|---|
| 100 MHz | 60 dB |
| 200 MHz | 60 dB |
| 400 MHz | 60 dB |
| 600 MHz | 60 dB |
| 800 MHz | 60 dB |
| 1000 MHz | 60 dB |

Ⓑ TABLE II. Illuminated chromaticity limits and luminance for sunlight readable displays (type S).

| Color | Color code | X 1/ | Y 1/ | Minimum average character luminance with or without EMI/RFI shielding |
|---|---|---|---|---|
| Red | R | .695<br>.710<br>.655<br>.660 | .285<br>SL 2/<br>.325<br>SL 2/ | 150 foot-lamberts |
| Green | G | .300<br>.300<br>.380<br>.380 | SL 2/<br>.600<br>.600<br>SL 2/ | 200 foot-lamberts |
| Yellow | Y | .562<br>.570<br>.596<br>.605 | .415<br>SL 2/<br>.382<br>SL 2/ | 300 foot-lamberts |
| Blue | B | .230<br>.230<br>.320<br>.320 | .420<br>.350<br>.350<br>.420 | 150 foot-lamberts |
| White | W | .400<br>.460<br>.400<br>.460 | .420<br>.420<br>.380<br>.380 | 250 foot-lamberts |

1/ The chromaticities of the colors are expressed as "X" and "Y" coordinates on the CIE chromaticity diagram and are within the limits bound by the coordinates listed for each color.

2/ SL-spectrum locus (where intersected by other coordinate pair).



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Ⓑ TABLE III. Illuminated chromaticity limits and luminance for NVIS night vision goggle compatible, sunlight readable displays (type S).

| Color | Color code | u' 1/ | v' 1/ | r 1/ | Minimum average character luminance with or without EMI/RFI shielding |
|---|---|---|---|---|---|
| NVIS Green B | J | .131 | .623 | .057 | 200 foot-lamberts |
| NVIS Yellow | K | .274 | .622 | .083 | 200 foot-lamberts |

1/ Chromaticity is expressed as u' and v' coordinates of the 1976 UCS diagram. The values shown describe a circle whose center is at u', v', and of radius r. The area enclosed by this circle intersecting the spectral locus is the envelope limiting the acceptable color space.

Ⓑ TABLE IV. Illuminated chromaticity limits and luminance for visible legend displays (type N).

| Color | Color code | x 1/ | y 1/ | Character luminance with or without EMI/RFI shielding |
|---|---|---|---|---|
| BLUE/ WHITE | V | .400<br>.460<br>.400<br>.460 | .420<br>.420<br>.380<br>.380 | 0.5 to 3.0 foot-lamberts |
| RED | N | .695<br>.710<br>.655<br>.660 | .285<br>SL 2/<br>.325<br>SL 2/ | 0.5 to 3.0 foot-lamberts |

1/ Chromaticity is expressed as x and y on the CIE chromaticity diagram. Values shown are corners of the limiting envelope.

2/ SL-spectrum locus (where intersected by other coordinate pair).



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Ⓑ TABLE V. Illuminated chromaticity limits and luminance for NVIS night vision goggle compatible, visible legend displays (type N)

| Color | Color code | u' 1/ | v' 1/ | r 1/ | Character luminance with or without EMI/RFI shielding |
|---|---|---|---|---|---|
| NVIS Green A | M | .088 | .543 | .037 | 0.5 to 3.0 foot-lamberts |
| NVIS Green B | L | .131 | .623 | .057 | 0.5 to 3.0 foot-lamberts |

1/ Chromaticity is expressed as u' and v' coordinates of the 1976 UCS diagram. The values shown describe a circle whose center is at u', v', and of radius r. The area enclosed by this circle intersecting the spectral locus is the envelope limiting the acceptable color space.

Ⓑ TABLE VI. Minimum contrast for sunlight readable displays (type S) with or without EMI/RFI shielding. 1/

| Color 2/ | Color code | Condition 1 (ø1 = ø2 = 15°C) Contrast ratio | Condition 2 (ø1 = ø2 = 30°) Contrast ratio |
|---|---|---|---|
| RED | R | 0.6 | 0.2 |
| GREEN | G | 0.6 | 0.2 |
| YELLOW | Y | 0.6 | 0.3 |
| BLUE | B | 0.4 | 0.15 |
| WHITE | W | 0.6 | 0.15 |

1/ Minimum average contrast of each legend character.

2/ Application note: In addition to luminance contrast, color contrast is an important factor in readability. Colors in table are listed in order of general preference. Red has the best color contrast during sunlight conditions; Blue and white have the poorest color contrast.



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Ⓑ TABLE VII. Minimum contrast for NVIS night vision goggle sunlight read able displays (type S) with or without EMI/RFI shielding. 1/

| Color 2/ | Color code | Condition 1 (ø1 = ø2 = 15°C Contrast ratio |
|---|---|---|
| NVIS Green B | J | 0.4 |
| NVIS Yellow | K | 0.4 |

1/ Minimum average contrast of each legend character.

2/ Condition 2 is not applicable to NVIS compatible colors.

Ⓑ Part or Identifying Number (PIN): The PIN's for complete pushbutton switch assemblies are assigned as follows: 1/ 2/ 3/

M22885/90 - 7 2 P K 4

| 7 | 2 | P | K | 4 |
|---|---|---|---|---|
| Switch housing configuration (Table VIII) | Actuation and EMI/RFI shielding efficiency (Table IX) | Switch poles and contact material (Table X) | Lens module configuration and NVIS compatibility option (Table XI) | Lamp circuit and lens module style (Table XII) |
| Dripproof, split lamp ground, solder terminations | Alternate action without EMI/RFI shielding | Four pole with silver contacts | Horizontal split legend and NVIS compatible | Horizontal split lamp ground, dripproof, without EMI/RFI shield |

1/ PIN's generated do not include segment color (codes from table II, III, IV, or V), or segment legend. Segment numbers are defined in table XIII for reference only.

2/ PIN's generated do not include lamps. When lamps are to be included with the switch, it shall be so specified on the acquisiton document. Lamp types useable in the switches are defined in table XIV.

3/ PIN's for switch body portion of switch only are listed in table XV and are inactive for new design.



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Ⓑ TABLE VIII. Switch type, lamp circuit, and termination codes.

| Switch type | Lamp circuit | Termination | Code |
|---|---|---|---|
| TYPE I (Unsealed, individual mount) | Common ground | Solder | 2 |
| | Split ground | Solder | 5 |
| | Split ground | PCB pins | 6 |
| TYPE II (Unsealed, matrix mount) | Common ground | Solder | 3 |
| | Split ground | Solder | 1 |
| | Split ground | PCB pins | 9 |
| TYPE III (Sealed, individual mount) | Common ground | Solder | 4 |
| | Split ground | Solder | 7 |
| | Split ground | PCB pins | 8 |

Ⓑ TABLE IX. Pushbutton action and EMI/RFI shielding efficiency codes.

| Pushbutton action | Without EMI/RFI shielding | With EMI/RFI shielding |
|---|---|---|
| Momentary | 1 | 4 |
| Alternate | 2 | 5 |
| Indicator | 3 | 6 |

Ⓑ TABLE X. Switch poles and contact material codes.

| Switch poles | Contact material | |
|---|---|---|
| | Silver | Gold |
| SPDT | 1 | 4 |
| DPDT | 2 | 5 |
| 3PDT | 3 | 6 |
| 4PDT | P | R |
| Indicator | 0 | |



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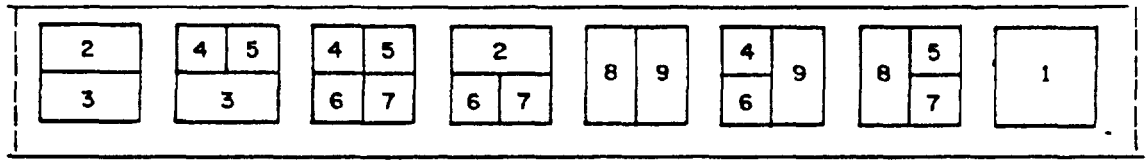
Ⓑ TABLE XI. Lens module configuration codes.

| | Configurations | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Standard lighting designations | B | C | D | E | F | G | J | H |
| NVIS compatible designations | K | L | M | N | P | Q | R | S |

Ⓑ TABLE XII. Lamp circuit and lens module style.

| Lamp circuit | Without EMI/RFI option | | With EMI/RFI option | |
|---|---|---|---|---|
| | Unsealed | Dripproof | Sealed | Dripproof |
| Common ground | 1 | 2 | L | M |
| Horizontal split ground | 3 | 4 | N | R |
| Vertical split ground | 5 | 6 | S | T |

Ⓑ TABLE XIII. Lens module legend positions (for reference only). 1/




1/ Segment numbers are to be used to identify color and legend positions.

Ⓑ TABLE XIV. Lamp types usable in the lens modules (for reference only). 1/

| Lamp type | MS number | Voltage | Current | MSCP | Life hours |
|---|---|---|---|---|---|
| 1 | --- | 5 | .060 | .15 | 6,500 |
| 2 | MS-24515 | 5 | .115 | .15 | 40,000 |
| 3 | --- | 12 | .040 | .15 | 16,000 |
| 4 | --- | 14 | .040 | .15 | 16,000 |
| 5 | --- | 18 | .026 | .15 | 10,000 |
| 6 | MS-3338 | 28 | .024 | .15 | 16,000 |

1/ Lamps with a mean spherical candle power of .15 must be used to obtain the sunlight readability specified. Replacement lamps must be aged and selected to meet this requirement.



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Ⓑ TABLE XV. Part numbers for switch body only. (Inactive for new design).

| M22885/90-(code letter shown below) | | | | | | |
|---|---|---|---|---|---|---|
| Code letters | | | | | | |
| Type I | Type II | Type III | Circuit | Operation | Contact material | EMI shielding |
| A | G | N | 1PDT | Momentary | Silver | No |
| B | H | P | 2PDT | Momentary | Silver | No |
| C | J | Q | 3PDT | Momentary | Silver | No |
| D | K | R | 1PDT | Alternate | Silver | No |
| E | L | S | 2PDT | Alternate | Silver | No |
| F | M | T | 3PDT | Alternate | Silver | No |
| AA | GA | NA | 1PDT | Momentary | Gold | No |
| BA | HA | PA | 2PDT | Momentary | Gold | No |
| CA | JA | QA | 3PDT | Momentary | Gold | No |
| DA | KA | RA | 1PDT | Alternate | Gold | No |
| EA | LA | SA | 2PDT | Alternate | Gold | No |
| FA | MA | TA | 3PDT | Alternate | Gold | No |
| U | V | W | Indicator | --- | --- | No |
| AB | GB | NB | 1PDT | Momentary | Silver | Yes |
| BB | HB | PB | 2PDT | Momentary | Silver | Yes |
| CB | JB | QB | 3PDT | Momentary | Silver | Yes |
| DB | KB | RB | 1PDT | Alternate | Silver | Yes |
| EB | LB | SB | 2PDT | Alternate | Silver | Yes |
| FB | MB | TB | 3PDT | Alternate | Silver | Yes |
| AC | GC | NC | 1PDT | Momentary | Gold | Yes |
| BC | HC | PC | 2PDT | Momentary | Gold | Yes |
| CC | JC | QC | 3PDT | Momentary | Gold | Yes |
| DC | KC | RC | 1PDT | Alternate | Gold | Yes |
| EC | LC | SC | 2PDT | Alternate | Gold | Yes |
| FC | MC | TC | 3PDT | Alternate | Gold | Yes |
| UB | VB | WB | Indicator | --- | --- | Yes |

1/ Part number codes are shown for switch body portion only. Lens modules, with required legends, dripproof seals, or EMI shielding shall be acquired separately from a source listed on the qualified products list. These parts are inactive for new design.



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Ⓑ Qualification inspection. See table XVI for group submission requirements.

Ⓑ TABLE XVI. Qualification inspection, group submission.

| Test sample | Qualification inspection table of MIL-S-22885 | | Additional testing | Extent of approval 1/ |
|---|---|---|---|---|
| | Group | Number of samples | | |
| M22885/90-75PBR 2/ | I<br>II<br>III<br>VI 3/ 4/<br>VII 5/ 6/<br>VIII 7/ | 10<br>4 (from group I)<br>2 (from group I)<br>2 (from group I)<br>2 (from group I)<br>22 7/ | | Type III |
| M22885/90-213K1 2/ | I<br>II<br>VI 3/ 8/ | 6<br>4 (from group I)<br>2 (from group I) | | Type I 9/ |
| M22885/90-923B3 2/ | | | -2 samples-<br>Visual and mechanical examination<br>Solderability<br>Terminal strength<br>Shock<br>Dielectric withstanding voltage<br>Operating characteristics | Type II 9/ |

1/ Includes gold and silver contact switches when the basic sensitive switches are qualified to MIL-S-8805/4 or /109.

2/ Test legend shall be:

| REMOTE |
|---|
| ACT LOWT |

3/ Four MS24515-718 or equivalent lamps shall be used in the lens module during group VI tests.

4/ Test inductive dc load only.

5/ Twenty five cycles of removal and reinstallation of the seal shall be made during the mechanical endurance test.

6/ Mechanical endurance with dripproof seal: 5000 cycles at -44°C to -48°C absolute minimum; 35,000 cycles at +25 ±2°C; 10,000 cycles at 85°C minimum. Cycling rate shall not exceed 18 cycles per minute.

7/ Twenty two lens modules only (with or without dripproof seal), two for each of eleven colors and display types shall be tested for color, luminance and sunlight readability where applicable. A single separate switch body shall be provided so that lens modules can be individually energized. NVIS color and NVIS radiance measurements shall be taken where applicable.

8/ Test resistive dc load only.

9/ When type III switches have been qualified.



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Group A inspection: See table XVII.

Ⓑ TABLE XVII. Group A inspection.

| Inspection |
|---|
| Seal (when applicable) 1/ |
| Visual and mechanical examination |
| Operating characteristic |
| Dielectric withstanding voltage |
| Contact resistance 2/ |
| Sunlight readability 3/ |
| NVIS compatibility 4/ |

1/ External inspection only.

2/ Contact resistance measurements for gold contact switches shall be measured in accordance with switch contact resistance of MIL-S-22885, except the test current shall not exceed 10 milliamperes and the open-circuit test voltage shall not exceed 30 millivolts dc. Maximum contact resistance is 3 ohms.

3/ Test one switch per color and use these as visual standards. Visually inspect all switches in each lot against the appropriately colored standard switch. The switches used as visual standards shall be certified annually by performing the sunlight readability test.

4/ Test one switch each for NVIS green B and NVIS yellow in accordance with the requirements of MIL-L-85762. These units will serve as standards when viewed with an image intensifier. Visually inspect all NVIS compatible switches in each lot with an image intensifier against the appropriate standard switch. The switches used as visual standards shall be certified annually by performing the NVIS compatibility test.



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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AV, MI
Navy - AS
Air Force - 17, 99
DLA - ES

User activities:
Army - AR
Navy - MC, OS, SH, YD
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5930-1374)



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